

Phys 20AL Week 4: Mass and Spring

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1 Aim for Experiment

The experiment is to measure the spring constant of a spring (which is a simple harmonic oscillator). In addition, it'll also be focusing on the static relation between the mass and the spring stretch of a spring, together with how variation of mass and maximum amplitude would affect the time period of a spring's dynamic motion.

2 Experimental Setup

Here there are three different sets of experiments, where two of them uses one setup, while the last one uses the other.

2.1 Experiments Recording data Manually

For this setup, there are two experiments involved: Collecting data of the spring's stretch under static scenario, and collecting data of the object's period under dynamic scenario.

The equipments include: Stand, two clamps, spring (that can endure long enough stretch without explicit deformation, typically a stretch of near 1 m), meter stick, a hook of 50 g, timer, mass scale, and masses of one 50 g, two 100 g, and one 200 g.

The following is the steps for setup:

1. Setup the stand at the edge of the table.
2. Fix one of the clamp on the top of the stand, and clamp one end of the spring onto it.
3. Fix the other clamp near the middle of the stand, and clamp the meter stick vertically onto it.

2.2 Experiment Recording data with Detector

For this setup, there is only one experiment, which is using the electronic detector to record the dynamic motion of the simple harmonic oscillator.

The equipments are nearly the same as in **Section 2.1**, except now it requires one more clamp, an electronic detector, and a computer.

The setup steps mostly follow from the one in **Section 2.1**, just additionally one needs to use the extra clamp to fix the detector on the stand, and plugin the detector to the computer for data recording.

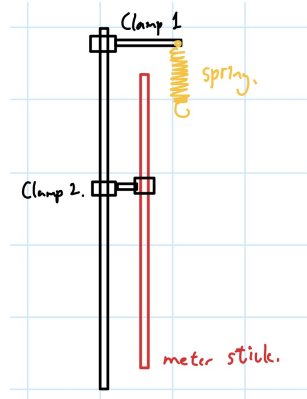


Figure 1: Sketch of the Setup with Manual Data Recording

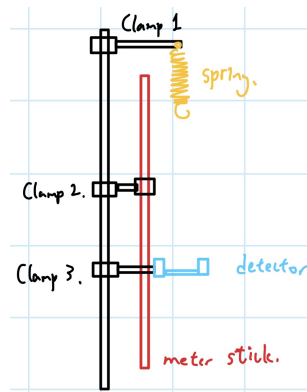


Figure 2: Sketch of the Setup with Manual Data Recording

3 Measurement & Method of Measure

In all the experiments, for any pairs of manipulated variables, we'll do a total of 3 trials each.

3.1 Static Approach

For the static approach, the aim is to measure when varying the mass (which corresponds to different gravitational force applied on the spring), how far the spring has stretched from original weightless scenario after the spring system reaches static equilibrium when the mass is added.

Here, the mass will be measured using a mass scale (unit: kg), and the stretch of the spring will be measured through the meter stick (unit: m). Which, the masses being tested are given by $n \cdot 50$ g (where integer n satisfies $1 \leq n \leq 10$). In terms of kg, it's given by $n \cdot 0.05$ kg, for every integer n satisfying $1 \leq n \leq 10$.

3.2 Dynamic Approach

For the dynamic approach, the aim is to measure the time period for the spring (i.e. a simple harmonic oscillator) to complete a cycle, and to observe how varying the mass and maximum amplitude of the spring system would affect this time period.

The mass measurement would again be done by a mass scale (unit: kg), and the maximum amplitude

of the oscillation will be measured through the meter stick (unit: m). The time period measurement then differ depending on the experiment:

- For the experiment recording data manually, we'll use a timer to measure the time period (unit: s).
- For the experiment recording data with detector, the electronic detector serve as an automated timer for measurement.

In both cases, we'll let the spring oscillates in total of 5 times, and record the total tie elapsed for the five cycles.

For both experiments, the proposed maximum amplitudes are 2, 4, 6 cm respectively (or 0.02, 0.04, 0.06 m).

For the mass that will be used, it differs by the experiment:

- For the experiment measuring time manually, the mass will be 100, 200, 300 g respectively (or 0.1, 0.2, 0.3 kg respectively).
- For the experiment measuring time with detector, the mass will be 100, 150, 200 g respectively (or 0.1, 0.15, 0.2 kg).

4 Experimental Procedure

4.1 Static Approach

We'll follow the procedure below:

1. Measure the initial static equilibrium point of the spring's free end using the meter stick (where meter stick should be fixed at the same height across the whole experiment).
2. Fix a mass that's going to be tested, put the mass on the hook, and hang the hook on the free end of the spring.
3. Wait till the spring system be at static equilibrium, then measure the new static equilibrium point of the spring's free end.
4. Record the difference of this new position away from the initial position recorded in Step 1.
5. Repeat Step 2 to 4 in total of three trials.
6. Change the mass to other ones awaited to be tested, and repeat Step 2 to 5 for each mass.

Rmk: any mass with the form $n \cdot 50$ g (where integer $1 \leq n \leq 10$) can be formed using two 50 g, two 100 g, and one 200 g, so the provided masses can complete the experiment.

4.2 Dynamic Approach

4.2.1 Experiment without Detector

We'll follow the procedure below:

1. Fix a mass that is being used, put the mass onto the hook, and hang it onto the spring until it reaches static equilibrium.

2. Fix a maximum amplitude that is being used (say l), pull the spring down until it is distance l away from the equilibrium point.
3. Release the spring, and start the timer simultaneously. Wait until the spring oscillates total of 5 cycles and record the time it takes to complete it.
4. Repeat Step 2 to 3 for a total of three trials.
5. Change the maximum amplitude to other values await to be tested, and repeat Step 2 to 4 for each desired maximum amplitude.
6. Change the mass to other values await to be tested, and repeat Step 2 to 5 for each mass.

4.2.2 Experiment with Detector

For the case with detector, it merely follows the procedure used in **Section 4.2.2**, except the timer's measurement is swapped using the detector.

As a remark, needs to make sure when starting the measurement, the detector is right on the equilibrium point (so if the mass varies, causing the equilibrium to change, need to adjust the height of the detector accordingly through fixing the clamp at corresponding height).

5 Experimental Data

5.1 Static Approach

Trial \ Mass (g)	50	100	150	200	250	300	350	400	450	500	Measure:	
1	4.500	9.500	14.800	19.600	24.600	29.600	34.600	39.700	44.700	49.800	Spring Displacement (cm)	
2	4.600	9.700	14.700	19.700	24.500	29.700	34.800	39.700	44.800	49.900		
3	4.700	9.600	14.800	19.800	24.700	29.500	34.700	39.800	44.800	49.700		

5.2 Dynamic Approach - Manual Recording

Mass (g):	100				
Equilibrium (cm):	25.5				
Trial \ Max Amplitude (cm):	2	4	6	Measure time (s)	
1	3.910	3.810	3.930		
2	3.780	3.890	3.920		
3	3.810	3.960	3.930		

Mass (g):	200				
Equilibrium (cm):	36				
Trial \ Max Amplitude (cm):	2	4	6	Time (s)	
1	4.980	4.930	5.020		
2	5.000	5.000	5.070		
3	5.030	5.000	5.030		

Mass (g):	300			
Equilibrium (cm):	46			
Trial \ Max Amplitude (cm):	2	4	6	
1	5.870	5.920	5.940	
2	5.930	5.890	5.940	
3	5.990	5.840	5.950	

5.3 Dynamic Approach - With Detector

mass (g):	100					
displacement:	2					
Trial 1		Trial 2		Trial 3		
200.219	0.064	286.004	0.059	296.570	0.052	
201.017	0.047	286.460	0.019	296.993	0.048	
201.460	0.043	286.789	0.052	297.353	0.050	
201.809	0.044	287.208	0.061	297.773	0.055	
202.227	0.054	287.574	0.053	298.138	0.053	
202.576	0.060	288.007	0.047	298.563	0.051	
203.008	0.055	288.361	0.053	298.928	0.047	
203.367	0.055	288.780	0.054	299.343	0.047	
203.795	0.065	289.142	0.025	299.705	0.053	
204.150	0.058	289.569	0.059	300.132	0.055	
204.608	0.021	289.933	0.054	300.497	0.051	
204.932	0.058					
Average Time						
0.878		0.786		0.785		

mass (g):	150					
amplitude (cm):	2					
Trial 1		Trial 2		Trial 3		
408.662	0.101	571.113	0.042	620.175	0.048	
409.095	0.102	571.542	0.087	620.644	0.044	
409.566	0.053	572.025	0.033	621.075	0.082	
410.050	0.052	572.455	0.044	621.549	0.044	
410.465	0.109	572.885	0.087	621.980	0.045	
410.904	0.035	573.348	0.040	622.415	0.080	
410.951	0.050	573.824	0.040	622.880	0.044	
411.368	0.098	574.249	0.087	623.351	0.044	
411.799	0.107	574.732	0.043	623.784	0.077	
412.274	0.065	575.163	0.038	624.252	0.049	
412.754	0.058	575.584	0.097	624.688	0.043	
413.173	0.102	576.055	0.044	625.118	0.081	
413.603	0.107	576.530	0.039	625.374	0.126	
414.073	0.055	576.959	0.085			
		579.422	0.193			
Average Period						
0.819		0.894		0.903		

mass (g):	200					
amplitude (cm):	2					
Trial 1		Trial 2		Trial 3		
730.660	0.101	835.202	0.095	1235.408	0.107	
731.259	0.087	835.744	0.105	1236.008	0.069	
731.667	0.144	836.209	0.094	1236.422	0.098	
732.262	0.083	836.754	0.094	1236.983	0.106	
732.720	0.048	837.212	0.097	1237.429	0.063	
733.219	0.147	837.781	0.067	1237.992	0.102	
733.686	0.089	838.222	0.095	1238.428	0.106	
734.273	0.091	838.765	0.097	1239.029	0.066	
734.683	0.152	839.230	0.066	1239.435	0.107	
735.281	0.047	839.770	0.098	1240.002	0.107	
735.707	0.089	840.231	0.100	1240.448	0.068	
735.815	0.021	840.776	0.096	1241.008	0.111	
736.235	0.154			1241.450	0.102	
736.714	0.077			1242.050	0.068	
737.272	0.335			1242.457	0.103	
				1243.025	0.101	
				1243.466	0.068	
Time						
1.009		1.006		1.008		